

Neural dynamics for working memory and evidence integration during olfactory navigation in *Drosophila*

Corresponding Author: Dr Katherine Nagel

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

This is a very interesting and clearly presented paper that provides direct evidence for neural integration of sensory inputs to guide behavior when direct contact with relevant cues is intermittent. This is supported by clever experiments to enable virtual replay of the experience of a fly following a turbulent odor plume. Of particular value is the demonstration not only of correlation between the relevant neural activity and behavior, but also the direct effects of interference with the neural activity on behavior, and modelling to support the behavioral relevance of the time course of neural activity that is observed. I have relatively few comments to make for improvement.

1) It might be helpful to expand on what are the implications of assuming the goal direction for olfactory steering is in allocentric coordinates (lines 47-48), in particular, what is the assumed reference frame. I found it unclear whether wind itself sets the reference frame (so the goal direction is defined relative to the wind, line 49) or whether it is assumed that other sensory inputs (e.g. ongoing proprioceptive estimation of changes in heading direction) determine the frame. Similarly, a clearer discussion of whether (and under what circumstances) the goal orientation is set to 'upwind' vs. some other goal direction with respect to the wind (or with respect to other directional cues) would clarify the mechanisms the authors have in mind.

2) Lines 95-97 "No systematic change in bump position when wind was rotated". If I understand this correctly, it suggests the wind itself is not providing the directional reference frame in which the animal is setting its goal? But then why does it change its heading? I note this is also problematic for the interpretation of the results in ref 28. This point is also relevant at lines 269-270 in the discussion.

3) Lines 201-212, figure 4, the silencing effects here, while apparently significant, do not seem clear cut, especially for the larger number of pulses. This could be commented on more clearly rather than simply concluding that the neurons labeled by VT062617 "contribute" to maintaining a trajectory – it would seem that other neurons might also contribute, as the behaviour is not fully abolished by silencing?

4) Paragraph starting line 296: The concepts here could be more clearly expressed. In particular, there seems to be a sudden jump into describing attractor networks without clearly linking this to the foregoing issue of persistent or integrative memory.

5) Line 272-3 "integration and persistence in these neurons appears to be probabilistic, rather than fixed" - could refer to the specific evidence that supports this. I'm not sure this has been mentioned in the discussion of the results, other than noting that the persistence is variable across flies/trials?

6) It might be also be useful to more clearly distinguish whether the data support encoding of a 'goal direction' vs. a 'location estimate' where the latter implies both direction and distance to the goal (represented as a vector). This is hinted at (lines 313-314) but not sufficiently explained. Perhaps it cannot be resolved in this study, but that itself would be worth explicit mention.

(Remarks on code availability)

I have checked that the code is available but not tested if it can be installed and run (it requires a Matlab license).

Reviewer #2

(Remarks to the Author)

This work by Kathman et al. reports an in-depth characterization of neurons labeled in VT062617 Gal4 line. They discovered that FB neurons in this line, potentially hΔK neurons, show ramping activity in response to repeated exposure to odor stimuli and that they maintain this activity after the off-set of the stimuli. These results appear to support the evidence-integration and working memory hypotheses. How the navigation system develops a goal direction and how it can be maintained have been big questions in the field. This work uses a highly original and advanced methodology to show that these FB neurons may perform these functions, which is a conceptually important progress in the field of navigation, providing a key substrate to understand the underlying circuit mechanisms.

I did not notice any significant issues in methodology and data analyses. Overall, the manuscript is very close to the final stage for publication, which I support enthusiastically. I have only a few minor comments.

The authors suggest that persistence arises from recurrent connectivity between hΔK and PFG neurons, but no direct evidence is provided. I know that the authors deposited a bioRxiv preprint addressing this issue. I understand the logic behind splitting the story into two papers, so I will not criticize it, but it would have been great if these two papers were one paper. Or, at least, I suggest citing the other paper to lead readers to a potential mechanism of the persistent activity.

Although hΔK neurons are implied to underlie the mechanism, other neurons in the VT62617 may also be involved in the phenomena described in this work. Does hΔK explain the all variance observed in neurons of VT62617? The reasoning or discussion behind excluding other possibilities of other types of neurons would strengthen the manuscript.

Ext. Data Fig. 1l and 1m seem very important results. I think it should be in the main figure and more analyses. There seem to be only five examples. Are these all? Are there more examples? What does this mean? If the wind direction changes, shouldn't the goal direction be corrected to represent the real goal direction? Does this mean that if the fly's body orientation is suddenly changed, the fly would ignore this change and walk straight?

Ext. Data. Fig. 1i: What each point means is unclear. Please clarify.

What is a boundary layer plume?

Fig. 1i and 2d: What are the blue shaded areas?

The $\tau_{\text{real}} = 7.8\text{s}$ in VT62617 line. How is this value calculated? Also, is the τ_{real} from hΔK data similar to this value?

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

This paper investigates how working memory and evidence integration, two core cognitive processes, are implemented in the central brain of *Drosophila* during olfactory navigation. Using a closed-loop virtual plume paradigm, calcium imaging, and optogenetic manipulations, the authors identify a specific population of fan-shaped body (FB) local neurons (labeled by VT062617) that exhibit persistent activity and gradual integration of odor cues. These dynamics correlate with maintaining a goal-directed heading after odor loss, a behavior critical for navigating turbulent odor plumes. Computational modeling further demonstrates that the persistence time constant observed in these neurons is near-optimal for plume navigation. The strengths and weaknesses of the study are described below

Strengths

1. The issue addressed is important

The manuscript tackles a fundamental question in neuroscience: how working memory and evidence integration are implemented in a compact neural circuit during naturalistic navigation. By focusing on *Drosophila*, the authors leverage a genetically tractable system to address computations that are typically studied in vertebrates, making this work highly relevant and impactful. The study also finds circuit evidence for some of the recent ideas in navigation in turbulent plumes.

2. Experiments and modeling are well-designed

The closed-loop virtual plume paradigm is an elegant and technically sophisticated approach that mimics the stochastic nature of odor plumes while allowing precise control over sensory input. This design enables the authors to probe neural dynamics under ecologically realistic conditions. Optogenetic silencing of VT062617 neurons provides strong causal support for their role in maintaining goal-directed trajectories after odor loss. This goes beyond correlation and establishes functional relevance. The computational model is well-conceived and provides useful information about how working memory is employed,

Weaknesses

1. The evidence for whether activity in the VT06261 neurons is correlated with bumps is weak. There is some evidence in figure 1i-j but it would be better to plot behavioral metric to compare directly with the physiology data in Figure 1f. Goal deviation or straightness on a trial-by-trial basis can be plotted and compared. It is fine if these metrics show that the relation

between physiology and behavior is weak. A separate point is how the straightness metric compares to behavior before the odor. That is, it is unlikely that all of the post-odor behavior is attributable to the bump. There could also be problems with straightness as a metric as it seems to hover around 1 and there might be some weirdness due to the bounded nature of straightness. Is paired t-test a good metric for 1h. Bottomline is that the data indicates that the authors' general conclusion seems to be supported, but better analyses are required to understand how much of the post-odor goal is signaled by these neurons.

2. While the experiments in Figure 2 are very difficult, both the evidence for whether the bump depends on odor frequency and evidence accumulation require more data/evidence. For Figure 2e, the number of flies is low, and the effect size could be problematic. There is a lot of variability across flies. A regression between instantaneous odor frequency and bump amplitude would give a better sense of how these quantities are related. Similarly, in Figure 2F, it is hard to assess how much of the bump amplitude is captured by odor history after controlling for the effects of locomotion.

3. Rationale for the local neurons is not clear

4. Effect size is small. The effect size should be discussed.

5. The basic locomotion characteristics of the model and flies are not matched as claimed by the authors. The difference between speeds and angular speed is huge. See between 0-5 mm/s data. The real flies seem to walk much faster. The same goes for the angular velocity. More importantly, the idea of an optimum integration window is not a good one, as the flies will not just walk past the odor. As they approach the odor, the signal will become continuous and a different navigational mode will take over. The authors can perhaps quantify the time to find the source.

6. Wind speeds used should be explicitly stated and discussed. The closed-loop paradigm uses a wind speed of 25 cm/s based on my calculation and their wind-tunnel experiment uses a speed of 12 cm/s. These wind speeds are reasonable, but a higher wind speed would reflect a more salient signal and might affect the accuracy and prevalence of upwind movement. Intuitively, if the windspeed is higher (without being too difficult to walk), the directional signal from air will be stronger and would dominate. The authors' choice of wind speed in different experiments should be explicitly stated, and confounds from the speeds being different should be discussed. For example, what is the straightness index during/post odor in the two conditions?

Minor: Some legends are unreadable – Example Figure 4f

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The revision fully addresses the points raised in my previous review, and strengthens an already excellent paper.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

All concerns were addressed.

(Remarks on code availability)

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REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

This is a very interesting and clearly presented paper that provides direct evidence for neural integration of sensory inputs to guide behavior when direct contact with relevant cues is intermittent. This is supported by clever experiments to enable virtual replay of the experience of a fly following a turbulent odor plume. Of particular value is the demonstration not only of correlation between the relevant neural activity and behavior, but also the direct effects of interference with the neural activity on behavior, and modelling to support the behavioral relevance of the time course of neural activity that is observed. I have relatively few comments to make for improvement.

We thank the reviewer for the positive comments on our manuscript!

1) It might be helpful to expand on what are the implications of assuming the goal direction for olfactory steering is in allocentric coordinates (lines 47-48), in particular, what is the assumed reference frame. I found it unclear whether wind itself sets the reference frame (so the goal direction is defined relative to the wind, line 49) or whether it is assumed that other sensory inputs (e.g. ongoing proprioceptive estimation of changes in heading direction) determine the frame. Similarly, a clearer discussion of whether (and under what circumstances) the goal orientation is set to 'upwind' vs. some other goal direction with respect to the wind (or with respect to other directional cues) would clarify the mechanisms the authors have in mind.

Thank you for this suggestion. We have tried to clarify our hypothesis arising from these findings (line 48-52): "Wind direction is known to drive rotation of compass neurons (EPGs, Okubo et al. 2020, May et al. 2025), and can be flexibly tethered to visual landmarks through plasticity in the compass system (Mussells Pires et al. 2024, Basnak et al. 2025). These experiments suggest that the fly may use the wind direction as an allocentric reference frame during odor-guided navigation, and that FB local neurons can encode goal directions relative to this reference frame."

2) Lines 95-97 "No systematic change in bump position when wind was rotated". If I understand this correctly, it suggests the wind itself is not providing the directional reference frame in which the animal is setting its goal? But then why does it change its heading? I note this is also problematic for the interpretation of the results in ref 28. This point is also relevant at lines 269-270 in the discussion.

This point was also raised by Reviewer 2. Our interpretation is that wind direction sets the external reference frame, since changes in wind direction are known to strongly drive changes in the position of the EPG bump, as shown in Okubo et al. If the VT062617 bump encodes a goal relative to this reference frame, it should not move when the wind is rotated, but the fly will still make a compensatory turn because there is now a mismatch between the goal direction and the reference frame. As these ideas seem important to the interpretation of the bump, we have moved these experiments to a main figure (new Figure 2), and expanded our discussion of these results in lines 119-128:

If activity bumps in VT062617 neurons represent a goal direction, we would expect them to remain stable even when the fly turns (Beetz et al. 2023, Mussells Pires et al. 2024). Consistent with this hypothesis, we observed little movement of the bump during turns, and poor correlation with heading (Fig. 2A,b). We would further expect that this bump should remain stable when the wind is rotated, which should drive a change in the fly's reference frame by rotating the representation in compass neurons (Okubo et al. 2020, Mussells Pires et al. 2024, Basnak et al 2025). When we rotated the wind during odor, we sometimes observed a compensatory change in heading, but this was not associated with any movement of the bump in VT062617 neurons (Fig. 2c,d). Compensatory turns could be driven by the difference

between the rotated wind-anchored reference frame and the stable goal signal representing a direction relative to the wind (Mussells Pires et al. 2024, Westeinde et al. 2024). We conclude that the activity bump in VT062617 most likely represents a short-term goal direction relative to wind that can be evoked by odor and persist after odor loss.

We have also slightly modified the Discussion where it references these experiments (line 292): "Rotation of the wind does not cause a rotation of the bump, consistent with the idea that it represents a goal direction relative to a reference frame anchored by wind, rather than a heading estimate."

3) Lines 201-212, figure 4, the silencing effects here, while apparently significant, do not seem clear cut, especially for the larger number of pulses. This could be commented on more clearly rather than simply concluding that the neurons labeled by VT062617 "contribute" to maintaining a trajectory – it would seem that other neurons might also contribute, as the behaviour is not fully abolished by silencing?

This is a reasonable point. We have added this consideration to line 226-228: "We conclude that FB local neurons labeled by VT062617 neurons contribute causally to maintaining a trajectory after odor loss, but that other neurons are likely to contribute to the initial selection of a goal trajectory."

4) Paragraph starting line 296: The concepts here could be more clearly expressed. In particular, there seems to be a sudden jump into describing attractor networks without clearly linking this to the foregoing issue of persistent or integrative memory.

Thank you for this feedback. This point was also made by another reviewer. We have updated the text in this paragraph to reference our follow-up manuscript which relates activity in hΔK neurons to their recurrent connectivity (line 326-328): "In a follow-up paper, we investigate the dynamics of these two populations using specific split-Gal4 lines, and show that recurrent connections between them are critical for the formation of persistent bump activity (Lanz et al. 2026)."

5) Line 272-3 "integration and persistence in these neurons appears to be probabilistic, rather than fixed" - could refer to the specific evidence that supports this. I'm not sure this has been mentioned in the discussion of the results, other than noting that the persistence is variable across flies/trials?

We are referring to the finding in Fig. 1 that the persistence time of these neurons after odor varies across trials. We have updated the text to clarify this point (line 295).

6) It might be also be useful to more clearly distinguish whether the data support encoding of a 'goal direction' vs. a 'location estimate' where the latter implies both direction and distance to the goal (represented as a vector). This is hinted at (lines 313-314) but not sufficiently explained. Perhaps it cannot be resolved in this study, but that itself would be worth explicit mention.

This is a good question. There is recent evidence from behavioral genetic studies that flies can learn the location of a goal relative to odor landmarks but it is not clear yet whether the signal we have observed in VT062617 neuron represents a location or a direction. We have added text to clarify this point (line 340-343).

Reviewer #1 (Remarks on code availability):

I have checked that the code is available but not tested if it can be installed and run (it requires a

Matlab license).

Reviewer #2 (Remarks to the Author):

This work by Kathman et al. reports an in-depth characterization of neurons labeled in VT062617 Gal4 line. They discovered that FB neurons in this line, potentially hΔK neurons, show ramping activity in response to repeated exposure to odor stimuli and that they maintain this activity after the off-set of the stimuli. These results appear to support the evidence-integration and working memory hypotheses. How the navigation system develops a goal direction and how it can be maintained have been big questions in the field. This work uses a highly original and advanced methodology to show that these FB neurons may perform these functions, which is a conceptually important progress in the field of navigation, providing a key substrate to understand the underlying circuit mechanisms.

I did not notice any significant issues in methodology and data analyses. Overall, the manuscript is very close to the final stage for publication, which I support enthusiastically. I have only a few minor comments.

We thank the reviewer for these supportive comments!

The authors suggest that persistence arises from recurrent connectivity between hΔK and PFG neurons, but no direct evidence is provided. I know that the authors deposited a bioRxiv preprint addressing this issue. I understand the logic behind splitting the story into two papers, so I will not criticize it, but it would have been great if these two papers were one paper. Or, at least, I suggest citing the other paper to lead readers to a potential mechanism of the persistent activity.

We have added a brief description and citation of the other manuscript to the Discussion (line 326-328): "In a follow-up paper, we investigate the dynamics of these two populations using specific split-Gal4 lines, and show that recurrent connections between them are critical for the formation of persistent bump activity (Lanz et al. 2026)." While we see the appeal of combining these stories into a single paper, we felt that two papers were required to cover both the original discovery of persistent goal-related bump activity and its ethological relevance (this paper) and the mechanistic dissection of circuit mechanisms supporting this activity (Lanz paper). The studies were led by two different first authors and we hope the papers reflect this.

Although hΔK neurons are implied to underlie the mechanism, other neurons in the VT62617 may also be involved in the phenomena described in this work. Does hΔK explain the all variance observed in neurons of VT62617? The reasoning or discussion behind excluding other possibilities of other types of neurons would strengthen the manuscript.

We did perform some imaging from the other cell type labeled by VT062617— hΔF— but did not detect any odor-evoked bump activity in those neurons. We therefore think the activity observed in VT062617 largely arises from hΔK. This is confirmed by imaging from an hΔK-specific split Gal4 line in the Lanz et al. manuscript.

Ext. Data Fig. 1l and 1m seem very important results. I think it should be in the main figure and more analyses. There seem to be only five examples. Are these all? Are there more examples? What does this mean? If the wind direction changes, shouldn't the goal direction be corrected to represent the real goal direction? Does this mean that if the fly's body orientation is suddenly changed, the fly would ignore this change and walk straight?

We agree that this is an important experiment, and have moved these panels back to the main figure (now Fig. 2). Our interpretation is that wind direction sets the reference frame, as it is known to

strongly drive rotations of the EPG bump, even in non-walking flies (Okubo et al. 2020, May et al. 2025). In our wind-rotation experiments, we expect that this reference frame changes, as would be expected from previous results, while we show here that a “goal direction” encoded by VT062617 neurons does not. A compensatory turn would be driven by the difference between the fixed goal encoded in VT062617 neurons and the shifted reference frame encoded by EPGs. We have clarified these ideas in lines 119-128:

“If activity bumps in VT062617 neurons represent a goal direction, we would expect them to remain stable even when the fly turns (Beetz et al. 2023, Mussells Pires et al. 2024). Consistent with this hypothesis, we observed little movement of the bump during turns, and poor correlation with heading (Fig. 2A,b). We would further expect that this bump should remain stable when the wind is rotated, which should drive a change in the fly’s reference frame by rotating the representation in compass neurons (Okubo et al. 2020, Mussells Pires et al. 2024, Basnak et al 2025). When we rotated the wind during odor, we sometimes observed a compensatory change in heading, but this was not associated with any movement of the bump in VT062617 neurons (Fig. 2c,d). We conclude that the activity bump in VT062617 represents a short-term goal direction relative to wind that can be evoked by odor and persist after odor loss.”

Ext. Data. Fig. 1i: What each point means is unclear. Please clarify.

We have revised this to clarify what is plotted in this panel (line 1136-1138):

i: Mean bump persistence distributions of individual flies are significantly different from one another (one-way ANOVA, $p = 0.004$, $n = 135$, $N = 14$). Each row shows all bump persistence times for a single fly, with the mean indicated by a black line. Flies are sorted along the y axis by mean duration.

Note that in this revision, we applied a more stringent and consistent bump-selection criterion for post-odor persistence analyses, including only bumps that began during odor and persisted beyond odor offset. This reduced the number of included bumps relative to the previous draft.

What is a boundary layer plume?

A boundary layer plume is a plume that develops near a surface such as the ground. Boundary layer plume have different statistical features from plumes that develop in free air. We have clarified this point in line 72-73.

Fig. 1i and 2d: What are the blue shaded areas?

These are the time periods analyzed in the following panels. We have clarified this in the figure legends.

The $\tau_{\text{real}} = 7.8\text{s}$ in VT62617 line. How is this value calculated? Also, is the τ_{real} from hΔK data similar to this value?

This value came from fitting an exponential curve to the distribution of bump persistence times after odor in Fig. 1f. We did not image from a specific hΔK split-Gal4 line in this manuscript. Note that in this revision, we applied a more stringent and consistent bump-selection criterion, using only bumps that started during the odor period and persisted after odor offset. This resulted in a revised estimate of τ of 5.59 s.

Reviewer #3 (Remarks to the Author):

This paper investigates how working memory and evidence integration, two core cognitive processes, are implemented in the central brain of *Drosophila* during olfactory navigation. Using a closed-loop virtual plume paradigm, calcium imaging, and optogenetic manipulations, the authors identify a specific population of fan-shaped body (FB) local neurons (labeled by VT062617) that exhibit persistent activity and gradual integration of odor cues. These dynamics correlate with maintaining a goal-directed heading after odor loss, a behavior critical for navigating turbulent odor plumes. Computational modeling further demonstrates that the persistence time constant observed in these neurons is near-optimal for plume navigation. The strengths and weaknesses of the study are described below

Strengths

1. The issue addressed is important

The manuscript tackles a fundamental question in neuroscience: how working memory and evidence integration are implemented in a compact neural circuit during naturalistic navigation. By focusing on *Drosophila*, the authors leverage a genetically tractable system to address computations that are typically studied in vertebrates, making this work highly relevant and impactful. The study also finds circuit evidence for some of the recent ideas in navigation in turbulent plumes.

2. Experiments and modeling are well-designed

The closed-loop virtual plume paradigm is an elegant and technically sophisticated approach that mimics the stochastic nature of odor plumes while allowing precise control over sensory input. This design enables the authors to probe neural dynamics under ecologically realistic conditions. Optogenetic silencing of VT062617 neurons provides strong causal support for their role in maintaining goal-directed trajectories after odor loss. This goes beyond correlation and establishes functional relevance. The computational model is well-conceived and provides useful information about how working memory is employed,

We thank the reviewer for these positive comments and notes on the strength of our work. We appreciate the weaknesses identified below and have worked to address these in this revision through additional analyses.

Weaknesses

1. The evidence for whether activity in the VT06261 neurons is correlated with bumps is weak. There is some evidence in figure 1i-j but it would be better to plot behavioral metric to compare directly with the physiology data in Figure 1f. Goal deviation or straightness on a trial-by-trial basis can be plotted and compared. It is fine if these metrics show that the relation between physiology and behavior is weak. A separate point is how the straightness metric compares to behavior before the odor. That is, it is unlikely that all of the post-odor behavior is attributable to the bump. There could also be problems with straightness as a metric as it seems to hover around 1 and there might be some weirdness due to the bounded nature of straightness. Is paired t-test a good metric for 1h. Bottomline is that the data indicates that the authors' general conclusion seems to be supported, but better analyses are required to understand how much of the post-odor goal is signaled by these neurons.

This is a reasonable point. Based on the reviewer's feedback we have conducted a new set of analysis looking at each detected bump and the characteristics of trajectories associated with these bumps compared to equivalent durations that precede each bump. Our new analyses (Fig. 1h and Ext

Data Fig. 1j) argue that the presence of a bump is associated with straighter trajectories, as measured by the standard deviation of heading and angular speed during the bump. In addition, we show a new panel (Fig. 1e) that illustrates bump timing for all bumps; showing that bumps are preferentially activated during odor but often persist into the post-odor period. We repeated the heading standard deviation analysis for bumps starting in the pre-odor, odor and post-odor periods and found that heading standard deviation was significantly reduced for bumps starting in both the pre-odor and odor periods. We did not observe a significant reduction for bumps starting in the post-odor period. We think this is likely because the "pre-bump" period for these falls during the odor. Based on these analyses we have updated Figure 1, Extended Data. Fig. 1 and our exposition of our findings (lines 95-117). Our main conclusions are:

- bump activity is preferentially activated by odor but can persist into the post-odor period
- when a bump occurs it is correlated with the fly adopting a straighter trajectory
- this means that bumps persisting after odor correlate with maintaining the trajectory adopted during odor and this is supported by our analysis of trajectories around bump offset

We think that the new analysis suggested by the reviewer clarifies the relationship between odor, bump activity, and features of trajectories. We note that in this revision, we applied a more stringent and consistent bump-selection criterion. For example, our offset analysis uses only bumps that started during the odor period and persisted after odor offset. This resulted in a revised estimate of τ of 5.59 s.

2. While the experiments in Figure 2 are very difficult, both the evidence for whether the bump depends on odor frequency and evidence accumulation require more data/evidence. For Figure 2e, the number of flies is low, and the effect size could be problematic. There is a lot of variability across flies. A regression between instantaneous odor frequency and bump amplitude would give a better sense of how these quantities are related. Similarly, in Figure 2F, it is hard to assess how much of the bump amplitude is captured by odor history after controlling for the effects of locomotion.

We appreciate this feedback as well. Based on the reviewer's feedback we examined correlations between bump amplitude and odor frequency on a bump-by-bump basis. This analysis did not support the idea that bumps encode odor frequency. We have therefore removed the analysis previously shown in 2E and the conclusion that bumps encode odor frequency. In contrast, we think that the evidence in favor of odor integration is strong. First, we observed a significant filter relating odor to bump amplitude in all 6 flies, all of which have an extended "integrating" shape (Supp Fig. 2). The duration of these filters is well-matched to the duration of persistent activity observed in Fig. 1. Because we have mathematically removed the influence of 1st -order correlations between locomotor parameters (forward velocity and angular speed) and odor, the presence of a significant filter argues that integrated odor input drives bump activity beyond that which is predicted by changes in locomotion alone. This interpretation is supported by our observation that we observe almost no bump activity in the absence of sensory stimuli (Ext Data Fig. 1e,f). We have clarified these points in the Results (lines 161-164): "This analysis allows us to examine the independent contribution of odor and behavior to bump activity, despite the fact that these variables are correlated in our dataset. Our decorrelated filter analysis reveals that both odor and behavior contribute to bump activity, with distinct dynamics."

3. Rationale for the local neurons is not clear

We assume this is a reference to Fig 3 of the original submission. We decided to include these imaging data from another broad local neuron type as they demonstrate that the same dynamics are not found everywhere and are specific to the line VT062617.

4. Effect size is small. The effect size should be discussed.

The effects size is large for the broader line —VT062617— and smaller for the specific ΔK split. We think this is expected given that split Gal4 lines often show weaker and more inconsistent expression. For example Mussells Pires et al. 2024 observed a non-significant reduction in orientation after silencing PFL3 neurons using a specific split-Gal4 line. We have added text (line 228) clarifying that we think VT062617 neurons work in concert with other local neuron to choose the initial goal direction, as the effect of silencing is primarily on persistence but not the early phase of the behavior.

5. The basic locomotion characteristics of the model and flies are not matched as claimed by the authors. The difference between speeds and angular speed is huge. See between 0-5 mm/s data. The real flies seem to walk much faster. The same goes for the angular velocity. More importantly, the idea of an optimum integration window is not a good one, as the flies will not just walk past the odor. As they approach the odor, the signal will become continuous and a different navigational mode will take over. The authors can perhaps quantify the time to find the source.

In Figure 6c we show the log probability distributions of forward and angular velocity for our measured walking ball data and our model. As the model distributions are constrained to be Gaussian (except for the peak around 0 corresponding to the stop state), we cannot match these distributions exactly. However, the means of the two distributions overlap to within 95% confidence intervals. We have added a representation of the means of each distribution along with confidence intervals. The discrepancy between real and model distributions at high speeds corresponds to very small probabilities.

We do not believe it is presently known what causes flies to stop when they reach an odor source. In our free-walking wind tunnel experiments we have observed that flies will continuously make upwind runs in response to continuous odor presentations for hours. It might be that other cues such as tastants drive stopping at food sources. We have added text with this caveat (line 257-259): "Although it is not known what causes flies to stop at a food source, our model suggests that very long integration times could be deleterious when searching a stochastic signal."

6. Wind speeds used should be explicitly stated and discussed. The closed-loop paradigm uses a wind speed of 25 cm/s based on my calculation and their wind-tunnel experiment uses a speed of 12 cm/s. These wind speeds are reasonable, but a higher wind speed would reflect a more salient signal and might affect the accuracy and prevalence of upwind movement. Intuitively, if the windspeed is higher (without being too difficult to walk), the directional signal from air will be stronger and would dominate. The authors' choice of wind speed in different experiments should be explicitly stated, and confounds from the speeds being different should be discussed. For example, what is the straightness index during/post odor in the two conditions?

We thank the reviewer for raising this point. We examined upwind velocity and the standard deviation of heading as a function of windspeed for the three speeds at which we had significant data on our walking ball: 8, 16, and 33 cm/s. This analysis suggests that upwind velocities do increase slightly, and heading deviations decrease, with windspeed over this range (Ext Data Fig. 4c). In our manuscript, we compare trajectories during bump and no-bump periods of the same trial (for walking ball imaging experiments) or between light and no-light trials (for wind tunnel experiments). We previously showed that windspeed is constant throughout trials in our wind tunnel paradigm (Alvarez-Salvado et al. 2018). We measured wind speed during our odor pulse in the walking ball paradigm and also found that it was constant (Ext Data Fig. 4d). Therefore, we do not think that using different wind speeds across trials, or between the walking ball and wind tunnel paradigms, can account for the bump correlations or effects of silencing that we report here.

We have added text to the section on free walking wind tunnel data to include this caveat (line 228-231):

We note that wind speeds in our freely walking wind tunnels (~12cm/s) are a subset of those we used on the walking ball, and that higher wind speeds promoted more effective upwind tracking (Ext. Data Fig. 4c). Wind speed was consistent throughout wind tunnel trials (Alvarez-Salvado et al. 2018) and during odor pulses on the walking ball (Ext Data Fig. 4d).

Minor: Some legends are unreadable – Example Figure 4f

We have moved all statistics in this figure to Extended Data Table 1 to improve readability.